

CS 5330 HW10

1. What is a residual connection and how does it change the function a particular layer of the network is learning?

A residual connection is where the input of the CNN is added to the output at the end of the convolution stack. This allows gradient descent to propagate back to a certain layer input much more strongly (with a gradient of 1), eliminating the vanishing gradient problem.

2. DenseNet uses skip connections rather than residual connections. How does the DenseNet architecture combine filters from different layers of the block?

The output at each layer is concatenated with the output at all subsequent layers

3. Within a DenseNet block, does DenseNet reduce the number of filters that need to be computed compared to an equivalent number of layers for a ResNet architecture? How does it do that?

Yes. The skip connections allow the gradients to flow back down the network and train the network effectively.

4. What is the difference between a residual connection and a skip connection?

A skip connection is where you take a input of a CNN layer and *concatenate* it to the output of the CNN block. A residual connection is where you *add* it to the output.

5. Why do residual and skip connections facilitate much deeper networks

They enable deeper networks because they speed up training. For the same training time, you can have a deeper network.

6. What is the special innovation that differentiates GoogleNet from other networks?

It included a component called a Inception Module, which pioneered this idea of splitting channels into multiple pathways made up of multi-scale features (low/high resolution/frequency filters), and reconnecting them later

7. How are 1×1 convolution layers similar to PCA?

1×1 convolution layers has a effect similar to doing an eigenvector projection, you have this input and you want to compress it by finding what components are most important, basically asking what components are the most important. A 1×1 conv layer might go from 256 channels to 64 channels. The layer itself isn't really finding any new patterns in the image, it is taking that input and condensing it down to a smaller version of itself. The weights of the layers clue the output in on which elements of the input are most important. Also, like PCA, these layers don't need a lot of computation.

8. Consider a standard CNN convolution stack for the ImageNet classification task. Features in the first layer integrate information from only the pixels inside of a first layer filter (e.g. a 7×7 filter, of stride 3).

- a. What about the second layer (e.g. a 3×3 filter of stride 1)?

This layer effectively integrates information from a $7 + (3-1) \times 3 = 13 \times 13$ window in the original image

- b. Is there a convolution layer where the feature being computed is integrating information from the entire image?

Yes, towards the end or at later convolutional layers, the entire image can definitely be integrated in to the layer.

9. What are the two major elements of a Transformer layer? What is the purpose of each?

Attention layer - transform tokens by comparing with other tokens (dot product). Tokens that are correlated have a high attention. Each token queries all the other tokens, asking for other tokens that are related. The dot product is taken with the key of each token it is compared to, and you see how good the fit is.

2 Linear Layers - fully connected to Transformed token. Allow the model to learn weights based on the attention layer output for each token individually (no mixing with other tokens). Allows better refining of each token

Also normalization layer (normalize all the activations across all tokens)

10. What is a multi-head transformer block? What is the purpose?

It passes the token through several different attention blocks (k, Q, V) (rather than just one) and gives several augmented token outputs. These token

outputs are concatenated and a learned matrix W is multiplied by it. This gives a new token of the same size as the input. It is important to track multiple attentions because a single word / token might have several connections to other words it needs to care about, and having multiple heads lets the model represent different independent attentions.